D D M M Y Y Y Y

1) The Probability distribution function of variate X is given by the following table:

X	0	1	2	3	4	5	6
P(X)	k	3k	5k	7k	9k	11k	13k

i) Find the value of k, ii) $P(x \geq 5)$ and iii) $P(3 < x \leq 6)$

Ans:- $\sum P(x_i) = 1$

i) $= k + 3k + 5k + 7k + 9k + 11k + 13k$

$= 49k = 1$

$k = 1/49$

x	0	1	2	3	4	5	6
P(x)	1/49	3/49	5/49	1/7	9/49	11/49	13/49

ii) $P(x \geq 5) = P(5) + P(6)$

$= 11/49 + 13/49 \Rightarrow 24/49$

iii) $P(3 < x \leq 6) = P(4) + P(5) + P(6)$

$= 9/49 + 11/49 + 13/49 \Rightarrow 33/49$

2) If the probability of a bad reaction from a certain injection is 0.001, determine the chance that more than two of 2000 individuals will have a bad reaction?

Ans:- The probability of bad reaction

$P = 0.001$

$n = 2000$

$\mu = M$

$\mu = np$

DD MM YYYY

$\mu = m = 2000 \times 0.001$

$m = 2$

we have poissons distribution

$P(x) = \frac{m^x e^{-m}}{x!} \Rightarrow \frac{2^x e^{-2}}{x!} = P(x)$

P(bad reaction more than 2)

$P(x > 2) = 1 - P(x \leq 2)$

$= 1 - [P(x=0) + P(x=1) + P(x=2)]$

$= 1 - \left[\frac{2^0 e^{-2}}{0!} + \frac{2^1 e^{-2}}{1!} + \frac{2^2 e^{-2}}{2!}\right]$

$= 1 - [0.135 + 0.270 + 0.270]$

$= 0.325$

3) Find the mean and standard deviation of poisson's distribution.

Ans:- Mean $\mu = \sum_{x=0}^{n} x \; P(x)$

$\mu = \sum_{x=0}^{\infty} x \cdot P(x)$

$\mu = \sum_{x=0}^{\infty} x \; \frac{m^x e^{-m}}{x!}$

$\mu = \sum_{x=0}^{\infty} x \; \frac{m^x e^{-m}}{x(x-1)!}$

$= \sum_{x=0}^{\infty} \frac{m \, m^{x-1} e^{-m}}{(x-1)!}$

$m^x = m \cdot m^{x-1}$
$x! = x(x-1)!$

D D M M Y Y Y Y

$$\mu = m \sum_{x=1}^{\infty} \frac{m^{x-1} e^{-m}}{(x-1)!}$$

$$\mu = m e^{-m} \sum_{x=1}^{\infty} \frac{m^{x-1}}{(x-1)!}$$

$$\mu = m e^{-m} \left[1 + \frac{m}{1!} + \frac{m^2}{2!} + \frac{m^3}{3!} + \cdots\right]$$

$$\mu = m e^{-m} e^{m}$$

$$\mu = m$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

Variance :-

$$\sum_{x=0}^{\infty} x^2 P(x) - \mu^2 \longrightarrow (1)$$

Consider, $\sum_{x=0}^{\infty} x^2 P(x)$

$$= \sum_{x=0}^{\infty} [x(x-1) + x] P(x)$$

$$= \sum_{x=0}^{\infty} x(x-1) P(x) + \sum_{x=0}^{\infty} x \cdot P(x)$$

$$= \sum_{x=0}^{\infty} x(x-1) P(x) + \mu$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{m^x e^{-m}}{x!} + m$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{m^x e^{-m}}{x(x-1)(x-2)!} + m$$

$$= \sum_{x=0}^{\infty} m^2 \frac{m^{x-2} e^{-m}}{(x-2)!} + m$$

$x! = x(x-1)(x-2)!$

$m^x = m^2 m^{x-2}$

DD MM YYYY

$$= m^2 e^{-m} \sum_{x=2}^{\infty} \frac{m^{x-2}}{(x-2)!} + m$$

$$= m^2 e^{-m} \left[1 + \frac{m}{1!} + \frac{m^2}{2!} + \frac{m^3}{3!} + \cdots\right] + m$$

$$\sum_{x=0}^{\infty} x^2 P(x) = m^2 e^{-m} e^{m} + m$$

$$\sum_{x=0}^{\infty} x^2 P(x) = m^2 + m$$

Use above result in eqn (1)

variance $= m^2 + m - m^2$

variance $V = m$

SD $= \sqrt{V}$

SD $= \sqrt{m}$

further we can say that the mean and variance are equal for the possion's distribution

4) The probability of a pen manufactured by a factor be defective is 1/10. If 12 such pens are manufactured, what is the probability that
i) exactly 2 are defective
ii) at least 2 are defective
iii) none of them are defective

Ans:- Let x be the number of defective

P(defective) $\Rightarrow$ $P = 1/10$

$P + q = 1$

$q = 9/10 \Rightarrow 0.9$

$n = 12$

W.K.T $P(x) = {}^{n}C_x\, p^x\, q^{n-x}$

$P(x) = {}^{12}C_x\, (0.1)^x\, (0.9)^{12-x}$

i) P(exactly 2 are defective)

$P(x=2) = {}^{12}C_2\, (0.1)^2\, (0.9)^{10}$

$P(x=2) = 0.2301$

ii) P(at least 2 are defective)

$x = 0, 1, 2, 3, 4 \dots 12$

$P(x \geq 2) = 1 - P(x < 2)$

$= 1 - [P(x=0) + P(x=1)]$

$P(x \geq 2) = 1 - [{}^{12}C_0\, (0.1)^0\, (0.9)^{12-0} + {}^{12}C_1\, (0.1)^1\, (0.9)^{12-1}$

$P(x \geq 2) = 0.3410$

$P(x \geq 2) = P(x=2) + P(x=3) + P(x=4) + P(x=5) = 1$

$\therefore P(x \geq 2) = 1 - P(x < 2)$

iii) P(none of them are defective)

$P(x=0) = {}^{12}C_0\, (0.1)^0\, (0.9)^{12-0}$

$P(x=0) = 0.2824$

5) Determine the value of k, so that the function $f(x) = k(x^2+4)$ for $x = 0, 1, 2, 3$ can serve as a probability distribution of the discrete random variable x: Also find i) $P(0 < x \leq 2)$ & ii) $P(x \geq 1)$

Ans:- Given the probability function for the discrete random variable $x = f(x) = k(x^2+4)$

$x = 0, 1, 2, 3$

x	0	1	2	3
f(x)	4K	5k	8k	13k

$k(0^2+4) = 4k$

$k(1^2+4) = 5k$

$k(2^2+4) = 8k$

$k(3^2+4) = 13k$

D D M M Y Y Y Y

W.K.T

$$\sum_{i=1}^{4} P(x = x_i) = 1$$

$$= 4k + 5k + 8k + 13k = 1$$

$$30k = 1$$

$$\boxed{k = 1/30}$$

i) $P(0 < x \leq 2)$

$P(0 < x \leq 2) = P(1) + P(2)$

$= 5k + 8k$

$= 13k$

$= 13/30$

ii) $P(x \geq 1) = ?$

$P(x \geq 1) = P(1) + P(2) + P(3)$

$= 5k + 8k + 13k = 26k$

$= 26/30$

6) Find the mean and standard deviation of Binomial distribution?

Ans:- Mean $\mu = \sum_{x=0}^{n} x \; P(x)$

$= \sum_{x=0}^{n} x \; {}^{n}C_x \; p^x \; q^{n-x}$

$= \sum_{x=0}^{n} x \; \frac{n!}{x!(n-x)!} \; p^x \; q^{n-x}$

$= \sum_{x=0}^{n} x \; \frac{n(n-1)!}{x(x-1)!(n-x)!} \; p^x \; q^{n-x}$

$$= \sum_{x=0}^{n} \frac{n(n-1)!}{(x-1)!(n-x)!} p^x q^{n-x}$$

$$= n \sum_{x=0} \frac{(n-1)!}{(x-1)!(n-1)-(x-1)]!} p \cdot p^{x-1} q^{(n-1)-(x-1)}$$

$$P(x) = \sum_{x=0}^{n} {}^{n}C_x \, p^x q^{n-x}$$

distribution

$$= np \sum_{x=1}^{n} {}^{n-1}C_{x-1} \, p^{x-1} q^{(n-1)-(x-1)}$$

$$= np \, (q+p)^{n-1}$$

$$= np \, (1)^{n-1}$$

$$\mu = np$$

The mean $\mu = np$

${}^{n}C_r = \frac{n!}{r!(n-r)!}$
$p^x = p \cdot p^{x-1}$
$n-x = (n-1)-(x-1)$
$x_1^2 = x(x-1)+x$

Variance :-

$$V = \sum_{x=0}^{n} x^2 P(x) - \mu^2$$

Consider

$$\sum_{x=0}^{n} x^2 P(x) = \sum_{x=0}^{n} [x(x-1)+x] P(x)$$

$$= \sum_{x=0}^{n} x(x-1) P(x) + \sum_{x=0}^{n} x P(x)$$

$$= \sum_{x=0}^{n} x(x-1) P(x) + \mu$$

$$= \sum_{x=0}^{n} x(x-1) \, {}^{n}C_x \, p^x q^{n-x} + np$$

$$= \sum_{x=0}^{n} x(x-1) \frac{n!}{x!(n-x)!} p^x q^{n-x} + np$$

$$= \sum_{x=0}^{n} \frac{x(x-1)\, n!}{x(x-1)(x-2)!\,(n-x)!}\, p^x q^{n-x} + np$$

Soln:

$$= \sum_{x=0}^{n} \frac{x(n-1)(n-2)!}{(x-2)!\,(n-x)!}\, p^x q^{n-x} + np$$

$$= x(n-1) \sum_{x=0}^{n} \frac{(n-2)!}{(x-2)!(n-2)-(x-2)!}\, p^2 p^{x-2} q^{(n-x)-(x-2)} + np$$

$$= n(n-1) p^2 \sum_{x=2}^{n} {}^{n-2}C_{x-2}\, p^{x-2} q^{(n-2)-(x-2)} + np$$

$$= n(n-1)p^2 (q+p)^{n-2} + np \qquad \Sigma p(x) = (q+p)^n$$

$$= n(n-1)p^2 + np$$

$$\sum_{x=0}^{n} x^2 p(x) = n^2p^2 - np^2 + np$$

∴ variance

$$V = n^2p^2 - np^2 + np - (np)^2$$

$$V = np - np^2$$

$$V = np(1-p)$$

$$V = npq$$

$$SD = \sqrt{V}$$

$$SD = \sqrt{npq}$$

$p+q = 1$
$q = 1-p$

7) A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives. Find the mean and variance of this distribution?

D D M M Y Y Y Y

Soln:- 2 microcomputers purchase out of 8 microcomputer can be ~~selected~~ selected in $8C_2$ ways

In 8 microcomputers 3 are defective and 5 are good

Let random variable x represents the number of defective microcomputers

$x = \{0, 1, 2\}$

$$P(X=0) = \frac{3C_0 \times 5C_2}{8C_2}$$

$$= \frac{5}{14}$$

$$P(x=1) = \frac{3C_1 \times 5C_1}{8C_2}$$

$$= \frac{15}{28}$$

$$P(x=2) = \frac{3C_2 \times 5C_0}{8C_2} \Rightarrow \frac{3}{28}$$

Probability distribution of x is as follows

$X = x_i$	0	1	2
$P(x_i)$	5/14	15/28	3/28

$$\text{Mean } (\mu) = \Sigma x_i P(x_i)$$

$$= (0)(5/14) + (1)(15/28) + (2)(3/28)$$

$$= 3/4$$

$$\text{variance } (V) = \Sigma x_i^2 P(x_i) - \mu^2$$

$= (0)^2 (5/14) + (1)^2 (15/28) + (2)^2 (3/28) - (3/4)^2$

$= \frac{45}{112}$

8) In a factory producing blades, the probability of any blade being defective is 0.002. If blades are supplied in packets of 10 using poisson distribution determine the number of packets containing

i) No defective

ii) one defective and

iii) Two defective blades respectively in a consignment of 10000 packets

Soln:-

$P = 0.002$

$n = 10$

$\mu = np = 10 (0.002) = 0.02$

$\therefore m = 0.02$

$P(x) = \frac{e^{-m} m^x}{x!}$

$P(x) = \frac{e^{-0.02} (0.02)^x}{x!}$

$f(x) = 10000 P(x)$

i) $P(0) = \frac{e^{-0.02} (0.02)^0}{0!} \Rightarrow 0.9802$

$f(0) = 10000 P(0) = 9802$ Packets

iii) $P(1) = \frac{e^{-0.02}(0.02)^1}{1!} \Rightarrow 0.0196$

$f(1) = 10000\,P(1) = 196$ packets

iii) $P(2) = \frac{e^{-0.02}(0.02)^2}{2!} \Rightarrow 0.0002$

$f(2) = 10000\,P(2) = 2$ packets

9) If the mileage (in thousands of miles) of a radial type is a random variable with exponential distribution with mean 40,000 miles. Determine the probability that the tyre will last

i) At less 20,000 miles

ii) At most 30,000 miles

Soln:- E.D given by $f(x) = \begin{cases} \alpha e^{-\alpha x} & \text{for } x>0 \\ 0 & \text{otherwise} \end{cases}$

Mean $\mu = 1/\alpha$

$\mu = 40,000$

$1/\alpha = 40,000$

$\alpha = 1/40,000$

$f(x) = \frac{1}{40,000} e^{-x/40,000}$

i) P(atleast 20,000 miles)

$P(x \geq 20,000)$

$\int_{20,000}^{\infty} f(x)\,dx = \int_{20,000}^{\infty} 1/40,000\; e^{-x/40,000}\,dx$

D D M M Y Y Y Y

$$= 1/40,000 \left[\frac{e^{-x/40,000}}{-1/40,000} \right]_{30,000}^{\infty}$$

$$= -[e^{-\infty} - e^{-1/2}]$$

$$= 0.6065$$

ii) P(atmost 30,000 miles)

$$P(x \leq 30,000)$$

$$\int_0^{30,000} \frac{1}{40,000} e^{-x/40,000} \, dx$$

$$= 1/40,000 \left[\frac{e^{-x/40,000}}{-1/40,000} \right]_0^{30,000}$$

$$= -[e^{-0} - e^{\frac{-30,000}{40,000}}]$$

$$= -[e^{-0.75} - e^{0}]$$

$$= 0.5276$$

10) The density function of a random variable X is given by

$$f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

i) Find k

ii) Find the cdf F(x) and use it to evaluate $P[0.3 < x < 0.6]$

Soln:- N.E.T

C.P.F

D D M M Y Y Y Y

$$f(x) = \begin{cases} k\sqrt{x} & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

i) $f(x \geq 0)$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^1 k\sqrt{x}\,dx = 1$$

$$k\int_0^1 2/3\,dx = 1$$

$$k = 2/3\,[x^{2/3}]_0^1 = 1$$

$$k = 2/3\,[1^{2/3} - 0^{2/3}] = 1$$

$$k = 2/3\,[1-0]$$

$$k = 3/2$$

$\sqrt{x} = 2/3$

ii) $P(0.3 < x < 0.6)$

$$\int_{0.3}^{0.6} k\sqrt{x}$$

$$= 3/2\,[2/3\,x^{3/2}]_{0.3}^{0.6}$$

$$= 3/2 \cdot 2/3\,[(0.6)^{3/2} - (0.3)^{3/2}]$$

$$= [0.4647 - 0.1643]$$

$$= 0.3004$$

D D M M Y Y Y Y

10) In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life 2040 hours and standard deviation of 60 hours. Estimate the number of bulbs likely to burn for
i) more than 2150 hours
ii) less than 1950 hours
iii) In b/w 1920 & 2160 hours

Sol^n:-

$\mu = 2040$ hours

$\sigma = 60$ hours

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{x - 2040}{60}$$

If $x = 2150$;

$z = 1.83$

If $x = 1950$, $z = -1.5$

If $x = 1920$, $z = -2$

If $x = 2160$, $z = 2$

$P(z > a) = 0.5 - \phi(a)$
$P(z < -a) = 0.5 - \phi(a)$
$P(-a < z < a) = \phi(a) + \phi(a$

i) $P(x > 2150) = P(z > 1.83)$

$= 0.5 - \phi(1.83)$

$= 0.5 - 0.4664$

$= 0.0336$

∴ The number of bulbs likely to burn for more than 2150 hours $= 2000 \times 0.0336$

$= 67.6$

$\simeq$ 67 bulbs

D D M M Y Y Y Y

ii) $P(x < 1950) = P(z < -1.5)$

$= 0.5 - \phi(1.5)$

$= 0.5 - 0.4332$

$= 0.0668$

∴ The no of bulbs likely to burn for less than 1950 hours $= 2000 \times 0.0668$

$= 133.6$

≈ 134 bulbs

iii) $P(1920 \leq x < 2160) = P(-2 < x < 2)$

$= \phi(2) + \phi(2)$

$= 2\phi(2)$

$= 2(0.4772)$

$= 0.9544$

The no of bulbs likely to burn for in b/w 1920 to 2160 hours

$= 2000 \times 0.9544$

$= 1908.8$

≈ 1909 bulbs

12) A random variable X has the following probability function for various of values of x

x	0	1	2	3	4	5	6	7
P(x)	0	k	2k	3k	3k	k^2	$2k^2$	$7k^2+k$

i) Find the value of k

ii) Evaluate $P[X < 6]$, $P(0 < X < 5)$, and $P[x \geq 6]$

D D M M Y Y Y Y

Soln:-

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

We must have $P(x) \geq 0$ and $\sum P(x) = 1$

for $k \geq 0$

Now $\sum P(x) = 1$

$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$9k + 10k^2 = 1$

$10k^2 + 9k = 1$

$k = 1/10$ and $k = -1$

i) $k = 1/10$

ii) $P(x < 6) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$

$= 0 + k + 2k + 2k + 3k + k^2$

$= 8k + k^2$

$= 8/10 + (1/10)^2$

$= 8/10 + \frac{1}{100}$

$= \frac{81}{100}$

$P(x \geq 6) = P(6) + P(7)$

$= 2k^2 + 7k^2 + k$

$= 2(1/10)^2 + 7(1/10)^2 + 1/10$

$= 91/100$

$P(0 < x < 5) = P(1) + P(2) + P(3) + P(4)$

$= k + 2k + 2k + 3k$

$= 8k$

$= 8 \times 1/10$

$= 8/10$

D D M M Y Y Y Y

15) In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. what is the probability that a shower will last for?
i) 10 minutes or more
ii) less than 10 minutes
iii) b/w 10 & 12 minutes

Soln:- we have,

$f(x) = \alpha e^{-\alpha x}, \quad 0 < x < \infty$

$M = 5$ minutes

Mean $= 5 = 1/\alpha$

$\alpha = 1/5$

$f(x) = 1/5 \; e^{-1/5 x}$

i) f (10 min (or) more)

$$\int_{10}^{\infty} f(x)\,dx = \int_{10}^{\infty} 1/5 \; e^{-1/5 x}\,dx$$

$$= 1/5 \left[\frac{e^{-1/5 x}}{-1/5} \right]_{10}^{\infty}$$

$$= -\left[e^{-\infty} - e^{-10/5}\right]$$

$$= -\left[0 - e^{-2}\right]$$

$$= 0.1353$$

ii) f (less than 10 min)

$$\int_{0}^{10} f(x)\,dx = \int_{0}^{10} 1/5 \; e^{-1/5 x}\,dx$$

$$= 1/5 \left[\frac{e^{-1/5} x}{-1/5} \right]_0^{10}$$

$$= - [e^{-10/5} - e^0]$$

$$= - [e^{-2} - 1]$$

$$= 0.8646$$

iii) f(b/w 10 & 12 min)

$$\int_{10}^{12} f(x)\, dx = \int_{10}^{12} 1/5\ e^{-1/5} x\ dx$$

$$= 1/5 \left[\frac{e^{-1/5} x}{-1/5} \right]_{10}^{12}$$

$$= - [e^{-12/5} - e^{-10/5}]$$

$$= - [0.0907 - 0.1353]$$

$$= 0.04446$$

4) out of 800 families with 5 children each, how many would you expect to have

i) 3 boys

ii) At least one boy

iii) At most two boys, assuming equal probabilities for boys and girls.

Sol^n: The probabilities for boys & girls are equal

let then p = probability of boys $p = 0.5$

q = probability of girls $q = 0.5$

$$\boxed{n = 5}$$

D D M M Y Y Y Y

W.K.T $P(x) = {}^nC_x\, p^x\, q^{n-x}$

$P(x) = {}^5C_x\, (0.5)^x\, (0.5)^{5-x}$

$P(x) = {}^5C_x\, (0.5)^5$

$f(x)$ Probability of 5 children out of 800 families

$f(x) = 800\, P(x)$

$f(x) = 800 \times {}^5C_x\, (1/2)^5$

$= 800 \dfrac{1}{32}\, {}^5C_x$

$f(x) = 25 \times {}^5C_x$

i) P (3 boys)

$P(x=3) = 2 \times {}^5C_3$

$= 250$

ii) $F(x \geq 1)$

$F(x) = 800 \times P(x)$

$F(x \geq 1) = 800 \times P(x \geq 1)$

$= 800\, [1 - P(x < 1)]$

$= 800\, [1 - 1/32\ {}^5C_0]$

$= 775$

iii) $F(x \leq 2)$

$= 25\, [{}^5C_0 + {}^5C_1 + {}^5C_2]$

$= 400$

(or)

$F(x \leq 2) = 800 \times P(x \leq 2)$

$= 800\, [1/32\, ({}^5C_0 + {}^5C_1 + {}^5C_2)]$

$= 400$

15) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and variance of the distribution

Soln:-

$P(x < 45) = 31\% = 0.31$

$P(x > 64) = 8\% = 0.08$

We have,

$$z = \frac{x-\mu}{\sigma}$$

If $x = 45$, $z = \frac{45-\mu}{\sigma} = z_1$ (say) $\rightarrow$ (1)

If $x = 64$, $z = \frac{64-\mu}{\sigma} = z_2$ (say) $\rightarrow$ (2)

$P(x < 45) = 0.31$

$P(z < z_1) = 0.31$

$0.5 + \phi(z_1) = 0.31$

$\phi(z_1) = 0.31 - 0.5$

$\phi(z_1) = -0.19$

| $P(z \geq a) = 0.5 + \phi(a)$

From table, $\phi(0.5) = 0.1915 \simeq 0.19$

$\therefore \phi(z_1) = -\phi(0.5) = \phi(-0.5)$

$\therefore z_1 = -0.5$

eqn (1) become

$$\frac{45-\mu}{\sigma} = -0.5$$

$45 - \mu = -0.5\sigma$

$45 = -0.5\sigma + \mu$ $\rightarrow$ (3)

NOW,

$P(x > 64) = 0.08$

D D M M Y Y Y Y

$P(z > z_2) = 0.08$

$0.5 - \phi(z_2) = 0.08$

$0.5 - 0.08 = \phi(z_2)$

$0.42 = \phi(z_2)$

$\therefore \phi(z_2) = 0.42$

| $P(z > a) = 0.5 - \phi(a)$

from table, $\phi(1.4) = 0.41924 \approx 0.42$

$\phi(z_2) = \phi(1.4)$

$z_2 = 1.4$

eqn (2) becomes

$$\frac{64 - \mu}{\sigma} = 1.4$$

$64 - \mu = 1.4\sigma$

$64 = 1.4\sigma + \mu \longrightarrow (4)$

on solving 3 & 4, we get

$\sigma = 10$, $\mu = 50$

16) Out of 800 families with 4 children each, how many families would be expected to have

i) 2 boys & 2 girls

ii) atleast one boy

iii) at most 2 girls. Assume equal probabilities for boys and girls

Soln

$N = 800$

$n = 4$

Let x be the number of boy children in a families

D D M M Y Y Y Y

$p = 1/2$, $q = 1 - p = 1 - 1/2 = 1/2$

$\therefore$ X follows binomial distribution

$P(X = x) = {}^{n}C_x \, p^x q^{n-x}$, $x = 0, 1, 2, 3, 4$

$\therefore P(X = x) = {}^{4}C_x (1/2)^x (1/2)^{4-x}$

$= {}^{4}C_x (1/2)^{x+4-x}$

$P(x = x) = \frac{{}^{4}C_x}{16}$ //

i) $P(2 \text{ boys } \& \text{ 2 girls}) = P(2 \text{ boys})$

$= P(x = 2)$

$= \frac{{}^{4}C_2}{16} = \frac{6}{16} = \frac{3}{8}$

Expected no of families with 2b & 2g children

$= 800 \times \frac{3}{8} = 300$

ii) $P(\text{atleast 1 boy}) = P(x \geq 1)$

$= P(x=1) + P(x=2)$

$= 1 - P(x < 1)$

$= 1 - P(x = 0) = 1 - 1/16 = 15/16$

Expected no of families with atleast one boy

$= 800 \times \frac{15}{16} = 750$

D D M M Y Y Y Y

iii) P(at most 2 girls) = P(x=2) + P(x=3) + P(x=4)

g: 0, 1, 2,

b(4, 3, 2)

$= \frac{4C_2}{16} + \frac{4C_3}{16} + \frac{4C_4}{16} = \frac{11}{16}$

∴ Expected no of families with at most 2 girls

$= \frac{11}{16} \times 800 = 550$

17) The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth

i) ends less than 5 minutes

ii) b/w 5 and 10 minutes

Soln:-

We have,

$f(x) = \alpha e^{-\alpha x}, \quad 0 < x < \infty$

Mean = $\mu = 1/\alpha = 5$

$\alpha = 1/5$

∴ $f(x) = 1/5\ e^{-x/5}, \quad 0 < x < \infty$

i) $P(x<5) = \int_0^5 f(x)\,dx$

$= \int_0^5 1/5\ e^{-x/5}\,dx$

$= 1/5 \left[\frac{e^{-x/5}}{-1/5} \right]_0^5$

$= -[e^{-x/5}]_0^5$

$= -[e^{-1} - e^{0}]$

$= -[0.3679 - 1]$

$= 0.6321$

ii) $P(5 < x < 10) = \int_5^{10} f(x)\,dx$

$= \int_5^{10} 1/5\ e^{-x/5}\,dx$

$= 1/5 \left[\frac{e^{-x/5}}{-1/5}\right]_5^{10}$

$= -[e^{-x/5}]_5^{10}$

$= -[e^{-10/5} - e^{-5/5}]$

$= -[e^{-2} - e^{-1}]$

$= -[0.1353 - 0.3679]$

$= 0.2326$

18) The probability distribution of a finite random variable X is given by

X	-2	-1	0	1	2	3
$P(x)$	0.1	k	0.2	$2k$	0.3	k

i) Find the value of k

ii) variance

iii) $P(X \leq 2)$

Soln:- We must have, $P(x) \geq 0$

$\therefore k \geq 0$

also we must have, $\sum P(x) = 1$

$0.1 + K + 0.2 + 2K + 0.3 + K = 1$

$4K + 0.6 = 1$

$4K = 1 - 0.6$

$4K = 0.4$

$K = 0.4/4 = 0.1$

The probability distribution is as follows

X	-2	-1	0	1	2	3
$P(x)$	0.1	0.1	0.2	0.2	0.3	0.1

$\mu = \Sigma x P(x)$

$= (-2)(0.1) + (-1)(0.1) + (0)(0.2) + (1)(0.2) + (2)(0.3) + (3)(0.1)$

$= 0.8$

Variance $(V) = \Sigma x^2 P(x) - \mu^2$

$V = (-2)^2 (0.1) + (-1)^2 (0.1) + (0)^2 (0.2) + (1)^2 (0.2) + (2)^2 (0.3) + (3)^2 (0.1) - (0.8)^2$

$V = 2.8 - (0.8)^2$

$V = 2.16$

$P(x \leq 2) = P(-2) + P(-1) + P(0) + P(1) + P(2)$

$= 0.1 + 0.1 + 0.2 + 0.2 + 0.3$

$= 0.9$

19) The number of accidents in a year to taxi drivers in a city follows a poisson distribution with mean 3. out of 1000 taxi drivers find approximately number of drivers with

i) more than 3 accidents in a year

ii) at most 2 accidents in a year

D D M M Y Y Y Y

Soln:- $\mu = m = 3$

$$P(x) = \frac{e^{-m} m^x}{x!}$$

$$P(x) = \frac{e^{-3} 3^x}{x!}$$

i) $P(X > 3) = 1 - P(0) - P(1) - P(2) - P(3)$

$= 1 - 0.0498 - 0.1494 - 0.2240 - 0.2240$

$= 0.3528$

The no of drivers with more than two accidents $= 1000 \times 0.3528 = 352.8$

353 drivers

ii) $P(x \leq 2) = P(0) + P(1) + P(2)$

$$= \frac{e^{-3} 3^0}{0!} + \frac{e^{-3} 3^1}{1!} + \frac{e^{-3} 3^2}{2!}$$

$= 0.0498 + 0.1494 + 0.2240$

$= 0.4232$

The no of drivers with atmost two accidents

$= 1000 \times 0.4232$

$= 423.2$

423 drivers

20) The marks of 1000 students in an exam follows normal distribution with mean 70 & standard deviation 5. Find the students whose marks will be

i) less than 65

ii) b/w 65 & 75. $A(1) = 0.3413$

Soln:-

Let x : marks of students

Given :-

$\mu = 70$

$\sigma = 5$

Now, $z = \frac{x - \mu}{\sigma}$

$z = \frac{x - 70}{5}$

i) Less than 65

$P(x < 65)$

If $x = 65$, $z = \frac{65 - 70}{5} = -1$

$\therefore P(z < -1) = P(z > 1)$

$= 0.5 - \phi(1)$

$= 0.5 - 0.3413$

$= 0.1587$

Marks of 1000 students $= 1000 \times 0.1587$

$= 158.7$

ii) b/w 65 & 75

$P(65 < x < 75)$

$\therefore P(-1 < z < 1) = 2\phi(1)$

$= 2(0.3413)$

$= 0.6826$

$\therefore$ No of students scoring b/w 65 & 75 marks

$= 1000 \times 0.6826$

$= 682.6$

$= 683$ students